CORTEX — State of the Art (SOA) Module

Module Specification v1.1 — February 2026

---

1. Overview

The SOA module transforms raw literature (from locked SLS sessions) into structured clinical evidence, performance benchmarks, and regulatory-grade narrative sections. It produces the foundational data that feeds the Clinical Evaluation Report (CER), the Validation Strategy, and the Benefit/Risk analysis.

A project can have multiple SOA analyses, each linked to one or more SLS sessions. The standard pattern is two tracks (Clinical SOA → Device SOA), but the architecture supports additional tracks (alternative treatments, specific subpopulations, etc.).

Three layers of output per SOA:

1. **Extraction Grids** — structured data extracted from each article
2. **Thematic Sections** — narrative content organized by MDR-aligned themes, with referenced claims
3. **Benchmarks & Comparison Tables** — pooled statistics and device comparisons (Device SOA)

---

2. SOA Analysis Types & Sequential Dependency

2.1 Types

Type	Purpose	Linked SLS Type	Key Output
clinical	Understand the pathology, clinical context, diagnostic pathways, unmet needs, identify similar devices	soa_clinical	Clinical landscape, endpoints that matter, similar device registry
similar_device	Benchmark competitor/equivalent device performance, build equivalence argumentation	soa_device	Performance benchmarks, claims, equivalence table, comparison tables
alternative	Assess alternative diagnostic/treatment methods	soa_clinical or ad_hoc	Context for benefit/risk analysis
custom	Any other analysis track	Any SLS type	User-defined

2.2 Sequential Dependency: Clinical → Device

The Clinical SOA must be completed (or at least Section 6 — Market Analysis finalized) before the Device SOA can be fully scoped. Section 6 of the Clinical SOA includes a similar device registry from which the user selects 1-2 devices. These define the scope of the Device SOA and its linked SLS session.

Enforcement in UI:

- When creating a `similar_device` SOA, the system checks if a clinical SOA exists on the project
- If the clinical SOA's Section 6 is not yet in `reviewed` or `final` status, a warning is shown
- The user can proceed anyway (not a hard block) but the warning is logged

---

3. Data Model

3.1 Hierarchy

```
┌──────────────────────────────────┐   ┌──────────────────────────────────┐
│   SOA Clinical                   │   │   SOA Similar Device             │
│                                  │   │                                  │
│ §1 Clinical Condition            │   │ §1 Equivalence Argument.         │
│ §2 Diagnostic Pathway            │   │ §2 Performance Claims            │
│ §3 Unmet Clinical Needs          │   │ §3 Clinical Benefit Claims       │
│ §4 Risk Analysis                 │ selects │ §4 Safety & Risk Claims      │
│ §5 Software & Technology         │ 1-2 devices │ §5 Limitations           │
│ §6 Market Analysis &             │──────────────▶│                        │
│   Similar Devices ⭐             │   │ + Benchmarks                     │
│                                  │   │ + Comparison Tables              │
└──────────────────────────────────┘   └──────────────────────────────────┘
```

```
Project
├─ SLS Session(s) [locked]
└─ SOA Analysis
   ├─ Configuration (type, linked SLS, grid template, section template)
   ├─ Grid Template (configurable extraction columns)
   ├─ Extraction Grids (1 row per included article)
   │  └─ Article Appraisal Record (pertinence, device classification, relevance)
   ├─ Quality Assessments (1 per article — dual system: regulatory + internal)
   ├─ Thematic Sections (ordered, with narrative + claims)
   │  └─ Claims (referenced to articles, tied to sections)
   ├─ Similar Device Registry — clinical SOA
   ├─ Benchmarks — device SOA
   └─ Comparison Tables — device SOA
```

3.2 SOA Analysis Record

Field	Description
id	UUID
project_id	Parent project
name	e.g., "Clinical SOA - Acute Ischemic Stroke"
type	clinical | similar_device | alternative | custom
linked_sls_ids	One or more locked SLS sessions feeding this analysis
status	draft → extraction → review → completed → locked
grid_template_id	Which grid template to use for extraction
section_template_id	Which section template defines the narrative structure
quality_tool_default	Default quality assessment tool (quadas2 | reading_grid_has | newcastle_ottawa | jbi | jadad)
parent_soa_id	For device SOA: reference to the clinical SOA it depends on (nullable)
notes	Free-form context notes

---

4. Thematic Sections

4.1 Concept

Each SOA analysis is organized into thematic sections — ordered narrative chapters that structure the analysis. Sections are the primary deliverable: they contain the regulatory-grade narrative text, supported by referenced claims and linked to extraction grid data.

Section templates define the default structure per SOA type. Users can customize (add, remove, rename, reorder sections) per analysis.

4.2 Section Record

Field	Description
id	UUID
soa_analysis_id	Parent SOA analysis
key	Machine-readable (e.g., clinical_condition.epidemiology)

Field	Description
title	Display title (e.g., "1.2 Epidemiology")
content	Rich text narrative (Markdown) — AI-generated + human-edited
source_article_ids	Articles referenced in this section
status	empty → drafted → reviewed → final
drafted_by	ai | manual
reviewed_by	User who reviewed
sort_order	Position in the SOA
parent_section_key	For sub-sections (e.g., `clinical_condition` is parent of `clinical_condition.epidemiology`)

4.3 Default Section Template: Clinical SOA

```
1. Clinical Condition
  ├── 1.1 Pathology Description
  │  Detailed description of the clinical condition the device addresses.
  │  Grid sources: all articles tagged with epidemiological/clinical data.
  │
  ├── 1.2 Epidemiology
  │  Incidence, prevalence, mortality, demographics, geographic distribution.
  │  Grid sources: population_size, incidence, prevalence, mortality_morbidity.
  │
  └── 1.3 Risk Factors & Prognosis
    Known risk factors, natural disease progression, prognostic factors.
    Grid sources: risk_factors.

2. Diagnostic Pathway
  ├── 2.1 Current Clinical Pathway / Standard of Care
  │  Step-by-step description of how patients are currently diagnosed and managed.
  │  Where does the device fit in this pathway?
  │  Grid sources: clinical_workflow.
  │
  ├── 2.2 Gold Standard (Reference Method)
  │  The accepted reference standard for diagnosis. Its strengths and limitations.
  │  Grid sources: reference_standard.
  │
  ├── 2.3 Alternative Diagnostic Methods
  │  Other diagnostic approaches beyond the gold standard.
  │  Grid sources: diagnostic_method.
  │
```

└── 2.4 Target Population & Subgroups
Characterization of the intended patient population.
Relevant subgroups (age, comorbidities, anatomical variants).
Grid sources: population_description.

3. Unmet Clinical Needs
├── 3.1 Limitations of Current Methods
│ What current diagnostic methods fail to do or do poorly.
│ Grid sources: limitations, unmet_needs.
│
├── 3.2 Gaps Identified in Literature
│ Explicit gaps noted by researchers in the field.
│
└── 3.3 Clinical Need for the Device
Justification: why does this device need to exist?
How does it address the identified unmet needs?

4. Risk Analysis
├── 4.1 Known Risks of the Diagnostic Approach
│ Risks inherent to the diagnostic method (e.g., radiation for CT, contrast agents).
│
├── 4.2 Risks of Diagnostic Errors
│ Clinical consequences of false positives and false negatives.
│ Grid sources: key_findings, limitations.
│
└── 4.3 Clinical Consequences of Errors
Impact on patient management, outcomes, and safety.

5. Software & Technology Landscape
├── 5.1 AI/ML in the Clinical Domain
│ Current state of AI adoption in this clinical field.
│ Level of maturity, acceptance by clinicians.
│
└── 5.2 Regulatory Landscape for SaMD
Relevant regulatory frameworks, guidance documents, standards.

6. Market Analysis & Similar Devices
├── 6.1 Market Overview
│ Existing commercial solutions in this space.
│ Grid sources: devices_mentioned, regulatory_status_mentioned.
│
├── 6.2 Similar Device Registry
│ Structured list of all devices identified (see Section 7 below).
│
└── 6.3 Selection of Most Similar Devices

Selection of 1-2 devices for equivalence analysis, with justification.
→ FEEDS DEVICE SOA

4.4 Default Section Template: Device SOA (Similar Devices)

```
1. Equivalence Argumentation
  ├── 1.1 Technical Equivalence
  │  Algorithm architecture, inputs (DICOM data, modality), outputs,
  │  processing platform, deployment model.
  │  Comparison: your device vs. similar device(s).
  │
  ├── 1.2 Biological Equivalence
  │  Same body site, same tissue interaction.
  │  For SaMD: typically "not applicable" or minimal.
  │
  ├── 1.3 Clinical Equivalence
  │  Same intended use, same clinical context, same target population,
  │  same clinical workflow integration.
  │
  ├── 1.4 Acceptability of Differences
  │  Where your device differs from the similar device.
  │  Argumentation for why each difference does not negatively
  │  impact safety or performance.
  │
  └── 1.5 Data Access Basis (MDR Article 61(5) Compliance)
    Documents the basis for accessing equivalent device data:
    - "Contractual access" or "Public data only"
    - Justification for sufficiency of public data
    - Acknowledgment of limitations

2. Performance Claims
  ├── 2.1 Performance Data from Literature
  │  Summary of performance data extracted from included articles.
  │  Grid sources: sensitivity, specificity, AUC, PPV, NPV, accuracy, dice_score.
  │
  ├── 2.2 Pooled Benchmarks
  │  Weighted averages with confidence intervals and heterogeneity analysis.
  │  Auto-generated from benchmark calculations (see Section 8).
  │
  ├── 2.3 Performance Claims
  │  Formal claim statements, each referencing supporting articles.
  │
  └── 2.4 Comparison Table
    Your device vs. similar devices vs. SOA benchmark (see Section 9).

3. Clinical Benefit Claims
```

```
  ├── 3.1 Demonstrated Clinical Benefits from Literature
  │   Grid sources: clinical_benefit, key_findings.
  │
  ├── 3.2 Benefits Specific to AI-Assisted Diagnosis
  │   Benefits unique to AI/software approach: consistency, availability (24/7),
  │   speed, quantitative measurements.
  │
  └── 3.3 Clinical Benefit Claims
      Formal claim statements referencing supporting articles.

4. Safety & Risk Claims
  ├── 4.1 Known Risks from Similar Devices
  │   Grid sources: adverse_events, false_positive_rate, false_negative_rate.
  │
  ├── 4.2 Adverse Events & Incidents from Databases
  │   Data from MAUDE (FDA), EUDAMED, manufacturer FSCAs.
  │
  ├── 4.3 Field Safety Corrective Actions
  │   Any FSCAs issued by manufacturers of similar devices.
  │
  └── 4.4 Risk Claims
      Formal risk-related claim statements referencing supporting evidence.

5. Limitations
  ├── 5.1 Limitations of the Evidence Base
  │   Quality of available evidence, gaps, biases.
  │   Grid sources: limitations, bias_concerns.
  │
  ├── 5.2 Gaps Identified
  │   Gaps that cannot be addressed by the current SOA.
  │   These feed directly into the PMS module as open gaps requiring PMCF actions.
  │
  └── 5.3 Limitation Claims
      Formal limitation statements with justification for acceptability.
```

4.5 Section Customization

- Users can add custom sections at any level (parent or sub-section)
- Users can hide/remove default sections (hidden sections retain their data)
- Users can rename section titles
- Users can reorder sections via drag & drop
- Custom section templates can be saved for reuse across projects

4.6 AI Section Drafting

For each section, the user can click "Draft this section":

1. Input to LLM: section title + description + relevant articles (filtered by grid sources mapping) + their extraction grid data + their digests + scope definition
2. Output: a Markdown narrative draft for the section, with inline article references formatted as `[Author, Year]`
3. The draft populates the section's content field — user edits in a rich text editor
4. Regenerate button available with optional user guidance ("focus more on epidemiology data", "make it more concise")
5. Section status moves from `empty` to `drafted`

4.7 Claims

Claims are formal statements tied to a section, each referencing one or more articles.

Claim record:

Field	Description
id	UUID
section_id	Parent section
claim_type	performance | clinical_benefit | safety | risk | limitation
claim_text	The claim statement
source_article_ids	Articles supporting this claim
source_benchmark_id	Benchmark this claim derives from (if applicable)
generated_by	ai | manual
is_edited	Whether the user modified the AI-generated text
status	draft | reviewed | final
sort_order	Position within the section

Claim generation:

- Performance claims are auto-generated from benchmarks (Section 8)
- Other claims are AI-generated during section drafting — the LLM identifies claimable statements and structures them with references

- All claims are editable and reviewable
- Claims are exportable as a standalone XLSX/DOCX claims list with article references

---

5. Grid Templates & Extraction

5.1 Grid Template System

Each SOA analysis uses a grid template that defines which data points to extract from each article. Templates are configurable per project — the user starts from a default template based on SOA type and can add, remove, rename, or reorder columns.

A grid template column has:

Property	Description
key	Machine-readable identifier (e.g., sensitivity, study_design)
label	Display name (e.g., "Sensitivity", "Study Design")
data_type	text | number | percentage | boolean | select | multi_select
options	For select/multi_select: list of allowed values
required	Whether this field must be filled for every article
poolable	Whether this field can be used in benchmark calculations (numbers/percentages only)
description	Help text / extraction guidance for the AI
sort_order	Display order in the grid
category	Grouping label (e.g., "Methodology", "Performance", "Safety")
section_mapping	Which thematic section(s) this column feeds

5.2 Default Template: Clinical SOA

Category: Study Identification

Column	Type	Poolable	Section Mapping
study_design	select (RCT, Cohort, Cross-sectional, Systematic Review, Meta-analysis, Guidelines, Case-Control, Case Series, Case Report, Literature Review, Other)	No	All
evidence_level	select (I, II, III, IV, V)	No	All
grade_of_recommendation	select (A, B, C, D, Expert Opinion, NA)	No	All
country_region	text	No	clinical_condition.epidemiology

Category: Epidemiology

Column	Type	Poolable	Section Mapping
population_size	number	Yes	clinical_condition.epidemiology
population_description	text	No	diagnostic_pathway.target_population
incidence	text	No	clinical_condition.epidemiology
prevalence	text	No	clinical_condition.epidemiology
mortality_morbidity	text	No	clinical_condition.epidemiology
risk_factors	text	No	clinical_condition.risk_factors

Category: Diagnostic Pathway

Column	Type	Poolable	Section Mapping
diagnostic_method	text	No	diagnostic_pathway.alternatives
reference_standard	text	No	diagnostic_pathway.gold_standard
clinical_workflow	text	No	diagnostic_pathway.standard_of_care
time_to_diagnosis	text	No	diagnostic_pathway.standard_of_care

Category: Outcomes & Gaps

Column	Type	Poolable	Section Mapping
key_findings	text	No	Multiple sections
clinical_benefit	text	No	unmet_needs.limitations
limitations	text	No	unmet_needs.limitations
unmet_needs	text	No	unmet_needs.gaps

Category: Market & Devices

Column	Type	Poolable	Section Mapping
devices_mentioned	text	No	market_analysis.overview
device_performance_reported	boolean	No	market_analysis.overview
regulatory_status_mentioned	text	No	market_analysis.overview

Category: Article Appraisal [NEW]

Column	Type	Poolable	Section Mapping
evaluated_device_type	select (subject_device, similar_device, other_device)	No	All
medical_condition_data	select (yes, no, new_indication)	No	clinical_condition
user_group	text	No	diagnostic_pathway
cohort_description	text	No	clinical_condition.epidemiology
nb_patients	number	Yes	clinical_condition.epidemiology
female_male_ratio	text	No	clinical_condition.epidemiology
mean_age	text	No	clinical_condition.epidemiology
condition_type_severity	text	No	clinical_condition
lost_to_followup	text	No	All
inclusion_criteria	text	No	diagnostic_pathway
exclusion_criteria	text	No	diagnostic_pathway
outcome_criteria	text	No	All
pertinence_type	multi_select (pivotal_data, establishment_sota, clinical_performance, clinical_benefit, safety_evaluation, biocompatibility, other)	No	All
relevance_to_intended_purpose	select (full_intended_purpose, specific_indication, all_patient_population, not_relevant)	No	All
pivotal_aspects_covered	multi_select (pivotal_performance, pivotal_safety, hazard_identification, risk_estimation_management, none)	No	All
other_aspects_covered	multi_select (benefit_risk_ratio, acceptability_side_effects, equivalence_determination, establishment_sota, none)	No	All
author_conclusions	text	No	All

5.3 Default Template: Device SOA (Similar Devices)

Category: Device Identification

Column	Type	Poolable	Section Mapping
device_name	text	No	equivalence
manufacturer	text	No	equivalence
regulatory_clearance	multi_select (CE, 510(k), De Novo, NMPA, None)	No	equivalence
algorithm_type	select (Deep Learning, Machine Learning, Rule-based, Hybrid, Unknown)	No	equivalence.technical
software_version	text	No	equivalence.technical

Category: Study Methodology

Column	Type	Poolable	Section Mapping
study_design	select (Retrospective, Prospective, Multi-center, Single-center, MRMC, Systematic Review, Meta-analysis, Other)	No	performance.data
evidence_level	select (I, II, III, IV, V)	No	All
grade_of_recommendation	select (A, B, C, D, Expert Opinion, NA)	No	All
dataset_size	number	Yes	performance.data
population_description	text	No	equivalence.clinical
scanner_types	text	No	equivalence.technical
reference_standard	text	No	performance.data

Category: Article Appraisal [NEW]

Column	Type	Poolable	Section Mapping
evaluated_device_type	select (subject_device, similar_device, other_device)	No	All
nb_patients	number	Yes	performance.data
female_male_ratio	text	No	performance.data
mean_age	text	No	performance.data
condition_type_severity	text	No	equivalence.clinical
inclusion_criteria	text	No	performance.data
exclusion_criteria	text	No	performance.data
outcome_criteria	text	No	performance.data
pertinence_type	multi_select (pivotal_data, establishment_sota, clinical_performance, clinical_benefit, safety_evaluation, other)	No	All
relevance_to_intended_purpose	select (full_intended_purpose, specific_indication, all_patient_population, not_relevant)	No	All
pivotal_aspects_covered	multi_select (pivotal_performance, pivotal_safety, hazard_identification, risk_estimation_management, none)	No	All
other_aspects_covered	multi_select (benefit_risk_ratio, acceptability_side_effects, equivalence_determination, establishment_sota, none)	No	All
author_conclusions	text	No	All

Category: Performance (all poolable)

Column	Type	Poolable	Section Mapping
sensitivity	percentage	Yes	performance.benchmarks
specificity	percentage	Yes	performance.benchmarks
auc	percentage	Yes	performance.benchmarks
ppv	percentage	Yes	performance.benchmarks
npv	percentage	Yes	performance.benchmarks
accuracy	percentage	Yes	performance.benchmarks
dice_score	percentage	Yes	performance.benchmarks
processing_time	text	No	performance.data

Category: Clinical Benefit

Column	Type	Poolable	Section Mapping
clinical_benefit_reported	text	No	clinical_benefit.literature
time_savings	text	No	clinical_benefit.ai_specific
workflow_improvement	text	No	clinical_benefit.ai_specific
patient_outcome_impact	text	No	clinical_benefit.literature

Category: Safety

Column	Type	Poolable	Section Mapping
false_positive_rate	percentage	Yes	safety.known_risks
false_negative_rate	percentage	Yes	safety.known_risks
adverse_events	text	No	safety.known_risks
limitations	text	No	limitations.evidence_base
bias_concerns	text	No	limitations.evidence_base

5.4 Template Customization

- Users can add custom columns (any data type) to any template
- Users can hide/show default columns (hidden ≠ deleted — data is preserved)
- Users can reorder columns within and across categories
- Users can rename column labels (the key stays stable for data integrity)
- Users can map columns to sections (which thematic section does this column feed)
- Custom templates can be saved as project templates for reuse
- Changes to a template after extraction has started: new columns are added empty, hidden columns keep their data

---

6. AI Extraction Workflow

6.0 Upstream Dependency: Clinical Evaluation Plan (CEP)

Each SOA analysis must be consistent with the Clinical Evaluation Plan (CEP) defined at project level. The CEP specifies the appraisal criteria, the quality assessment method (e.g., QUADAS-2 + internal reading grids), and which endpoints are critical. The SOA module references the CEP and warns if the analysis deviates from the planned methodology.

6.1 Process

For each included article from the linked SLS:

1. Input to LLM: article title + abstract + full-text PDF (if available) + grid template (column definitions with descriptions)
2. LLM extracts a value for each column in the template
3. Output: one grid row per article with all columns filled (or marked "not reported" / "not applicable")
4. Confidence indicators: for each extracted value, the AI provides a confidence level (high | medium | low) — low confidence values are visually flagged for human attention
5. Source quote: for key numeric values (performance metrics), the AI extracts the exact quote from the article where the value was found — stored for audit

6.2 Extraction Modes

Mode	Description	When to Use
Batch extraction	Run extraction on all articles with no grid data yet	Initial extraction
Single article	Extract/re-extract one article	After adding a new article or to regenerate

Mode	Description	When to Use
Column fill	Extract a single new column across all articles	After adding a column to the template

All modes are async with progress feedback and cancellable.

6.3 Human Review & Override

Grid view: a spreadsheet-like table with all articles as rows and grid columns as columns.

- Cells with low confidence are highlighted (amber background)
- Clicking a cell opens an edit panel showing:
  - AI-extracted value
  - AI confidence level
  - Source quote from the article (if available)
  - Link to open the PDF at the relevant section (if full-text available)
  - Edit field to override the value
  - Override reason (required when changing an AI-extracted value)
  - Original AI value is always preserved for audit trail
- Cells can be marked as "Not reported" or "Not applicable"

6.4 Grid Status per Article

Status	Meaning
pending	No extraction done yet
extracted	AI extraction complete, not reviewed
reviewed	Human has reviewed and confirmed/edited
flagged	Human flagged an issue — needs discussion

Progress indicator: "42/67 articles extracted, 28 reviewed"

6A. Methodological Quality Assessment (MEDDEV §8 Compliance)

6A.1 Purpose

MEDDEV 2.7/1 Rev 4 §8 requires that each piece of clinical data be appraised for its methodological quality and scientific validity. Notified Bodies specifically check for structured quality assessment. This is distinct from the extraction grid — it evaluates the study itself, not the data it reports.

6A.2 Dual Assessment System

CORTEX implements a **two-level quality assessment system** that satisfies both regulatory expectations and internal methodological rigor:

Level 1 — Regulatory Tool (internationally recognized)

Standardized tools expected by Notified Bodies and FDA reviewers. These generate the formal quality assessment included in the CER.

Study Type	Assessment Tool	Use Case
Diagnostic accuracy	QUADAS-2 (default)	Similar device studies, standalone performance studies
RCTs	Cochrane RoB 2 or Jadad Score	If RCTs are included (rare for SaMD)
Observational / Cohort	Newcastle-Ottawa Scale	Cohort studies in clinical SOA
Case series / reports	JBI Critical Appraisal Checklist	Case series, low-evidence studies

Level 2 — Internal Reading Grid (Avicenna methodology)

Detailed study-type-specific grids based on established appraisal methodologies (HAS / ANAES framework). These provide deeper methodological insight for the Clinical Affairs team's internal decision-making and are documented as supporting evidence.

Study Type	Reading Grid	Criteria Count
Randomized trial	Reading Grid — RCT	~25 criteria across 4 domains
Non-randomized trial	Reading Grid — MINORS	12 standardized criteria (0/1/2 scoring)
Cohort study	Reading Grid — Cohort	~20 criteria across 4 domains
Case-control study	Reading Grid — Case-Control	~20 criteria across 4 domains
Case series	Reading Grid — Case Series	~15 criteria across 4 domains
Literature review / Systematic review	Reading Grid — Literature Review	~15 criteria across 3 domains

Both levels are completed for every article. The system auto-suggests which tools to apply based on the `study_design` field from the extraction grid, but the user can override.

6A.3 Level 1 — QUADAS-2 (Adapted for Diagnostic Accuracy Studies)

Domain 1 — Patient Selection

Signal Question	Answer	Risk of Bias
Was a consecutive or random sample of patients enrolled?	Yes / No / Unclear	Low / High / Unclear
Was a case-control design avoided?	Yes / No / Unclear	
Did the study avoid inappropriate exclusions?	Yes / No / Unclear	
Applicability concern: Do the patients match the device's intended population?		Low / High / Unclear

Domain 2 — Index Test

Signal Question	Answer	Risk of Bias
Were the index test results interpreted without knowledge of the reference standard?	Yes / No / Unclear	Low / High / Unclear
Was the threshold/algorithm version pre-specified?	Yes / No / Unclear	
Applicability concern: Does the index test match the device under evaluation?		Low / High / Unclear

Domain 3 — Reference Standard

Signal Question	Answer	Risk of Bias
Is the reference standard likely to correctly classify the target condition?	Yes / No / Unclear	Low / High / Unclear
Were the reference standard results interpreted without knowledge of the index test?	Yes / No / Unclear	
Applicability concern: Does the reference standard match the gold standard defined in the clinical SOA?		Low / High / Unclear

Domain 4 — Flow and Timing

Signal Question	Answer	Risk of Bias
Was there an appropriate interval between index test and reference standard?	Yes / No / Unclear	Low / High / Unclear
Did all patients receive the same reference standard?	Yes / No / Unclear	
Were all patients included in the analysis?	Yes / No / Unclear	

Overall quality rating: high_quality | acceptable | low_quality — derived from domain scores.

6A.4 Level 2 — Internal Reading Grids (Full Specification)

Each reading grid follows a consistent structure:

Common header (all grid types):

This header captures the article-level appraisal data. It is pre-populated from the extraction grid where possible and completed/confirmed during quality assessment.

Field	Type	Pre-populated from
Publication information (title, authors, journal, year)	text	SLS article record
Purpose(s)	text	Extraction grid: key_findings
Study design	select	Extraction grid: study_design
Grade of recommendations	select (A, B, C, D, Expert Opinion, NA)	Extraction grid: grade_of_recommendation
Level of scientific evidence	select (I, II, III, IV, V)	Extraction grid: evidence_level
Statistics	text	Manual
Device information (product, characteristics, manufacturer)	text	Extraction grid: device_name, manufacturer
Evaluated device type	select (subject_device, similar_device, other_device)	Extraction grid: evaluated_device_type
Medical application / type of surgery	text	Extraction grid
Data concerning the medical conditions	select (yes, no, new_indication)	Extraction grid: medical_condition_data
User group	text	Extraction grid: user_group
Cohort (nb patients, F/M, mean age, condition severity)	composite	Extraction grid
Lost to follow-up	text	Extraction grid: lost_to_followup
Range of time (duration, repeat exposure, follow-up)	text	Manual
Conduct of the study	text	Manual
Inclusion / Exclusion criteria	text	Extraction grid
Indications / Contra-indications	text	Manual
Outcome criteria	text	Extraction grid: outcome_criteria
Results analysis	text	Manual
Pertinence of the study	multi_select	Extraction grid: pertinence_type
Relevance to intended purpose	select	Extraction grid: relevance_to_intended_purpose

Field	Type	Pre-populated from
Pivotal aspects covered	multi_select	Extraction grid: pivotal_aspects_covered
Other aspects covered	multi_select	Extraction grid: other_aspects_covered
Conclusion of the pertinence	text	Manual
Author conclusions	text	Extraction grid: author_conclusions

Methodological criteria scoring:

Each criterion in the reading grids is scored using a 5-option system:

Code	Meaning
Y	Yes — criterion is met
I	Incomplete — partially met
N	No — criterion is not met
NA	Not Applicable
DNK	Do Not Know — cannot determine from available information

Each criterion has an associated **Comments** field (free text).

Eliminatory criteria rule: Criteria marked in *italics* in the grid definitions below are **eliminatory**. If any eliminatory criterion receives a score of **N**, the study is automatically classified as **Unacceptable** regardless of other scores.

Overall quality conclusion (all grid types):

Rating	Description
Very good	Study meets nearly all criteria, no significant methodological concerns
Pretty good	Study meets most criteria, minor limitations acknowledged
Weak but acceptable	Study has notable limitations but provides usable data within its scope
Unacceptable	Study has critical methodological flaws — data should not be relied upon
Not qualified to judge	Insufficient information to assess quality

Grid Type 1: Randomized Trial

Domain	Criterion	Eliminatory
Study scheme	*Clear formulation of the objective*	✓
	A priori assumption	
	Compared interventions defined, justified and acceptable	✓
	Principle of ambivalence respected	✓
	Described and adequate draw	✓
	Blind draw by an independent centre	✓
Procedure of selection	*Inclusion criteria described and adequate*	✓
	Exclusion criteria described and adequate	
	Relevance of the indication	
	Indication of the number of refusals before the study	
	Identical procedures in all groups	
	Adapted sample size	
	Ethical rules respected	✓
Conduct of the study	*Description of the intervention methods sufficiently detailed*	✓
	Main outcomes defined and relevant	✓
	Reliable and valid measurement of expected effects	
	Blind evaluation of expected effects	
	Blind draw of subjects and caregivers	
	Withdrawals explained, balanced and reasonable (flow diagram)	
	Specified and sufficiently long follow-up time	✓
Results analysis	Comparability of groups at baseline presented in a table	
	Intent-to-treat (ITT) analysis	
	Taking into account monitoring (survival analysis)	
	Statistician's blind draw	
	Other interventions monitored and described	

Domain	Criterion	Eliminatory
	Side effects assessed and described	
	Enrolment monitored and described	
	Consideration of important variables	
	Verifiable raw data results	
	Taking into account multiple comparisons	
Other elements	Applicability outside the experimental context	
	Contribution of the intervention compared to the existing	

Grid Type 2: Non-Randomized Trial (MINORS Criteria)

Scored 0 (not reported), 1 (reported but inadequate), or 2 (reported and adequate). Global ideal score: 16 for non-comparative, 24 for comparative studies.

#	Criterion	Score (0/1/2)
1	Clearly stated aim	
2	Inclusion of consecutive patients	
3	Prospective collection of data	
4	Endpoints appropriate to the study aim	
5	Unbiased assessment of the study endpoint	
6	Appropriate follow-up period	
7	Loss to follow-up less than 5%	
8	Prospective calculation of the study size	
9	Adequate control group	
10	Contemporary groups	
11	Baseline equivalence	
12	Adequate statistical analysis	
Total	/16 (non-comparative) or /24 (comparative)	

Grid Type 3: Cohort Study

Domain	Criterion	Eliminatory
Study scheme	*Clear formulation of the objective*	✓
	A priori assumption	
	Description of the exhibitions	✓
	Acceptable reference groups	✓
Procedure of selection	*Inclusion criteria described and adequate*	✓
	Exclusion criteria described and adequate	
	Acceptable definition of start of monitoring	
	Independent disease risk selection	
	Reliable and valid measurement of input variables	
	Indication of the number of refusals before the study	
	Identical procedures in all groups	
	Adapted sample size	
	Ethical rules respected	
Conduct of the study	*Specified and sufficiently long follow-up time*	✓
	Same regular and standardized monitoring of all groups	
	Observed effects defined and relevant	✓
	Reliable and valid measurement of observed effects	✓
	Blind evaluation of observed effects	
	Withdrawals indicated, explained and reasonable	
	Balanced withdrawals between groups	
Results analysis	Comparison of groups at baseline	
	Taking into account monitoring (survival analysis)	
	Hidden exposures from the statistician	
	Other monitored and described exposures	
	Consideration of important variables	✓

Domain	Criterion	Eliminatory
	Verifiable raw data results	
	Taking into account multiple comparisons	
Other elements	Discussion of protocol violations	
	Discussion of causality criteria	

Grid Type 4: Case-Control Study

Domain	Criterion	Eliminatory
Study scheme	*Clear formulation of the objective*	✓
	A priori assumption	
	Description of the disease studied	✓
	Inclusion criteria described and adequate	✓
	Exclusion criteria described and adequate	
	Diagnostic dates and criteria clear and valid	
	Exclusion of disease in controls	
	Acceptable witness groups	✓
	Same original population for cases and controls	
	Independent exposure sampling	
	Indication of the number of refusals before the study	
	Identical procedures in all groups	
	Adapted sample size	
	Ethical rules respected	
Conduct of the study	*Defined and relevant exposures*	✓
	Acceptable exposure search period	✓
	Reliable and valid exposure measurement	✓
	Blind exposure measurement	
	Indicated, explained and reasonable withdrawals	
	Balanced withdrawals between groups	
Results analysis	Comparison of groups at baseline	
	Association measure adapted to the study scheme	
	Other exposures measured and described	
	Consideration of important variables	✓
	Verifiable raw data results	

Domain	Criterion	Eliminatory
	Taking into account multiple comparisons	
Other elements	Discussion of selection and measurement biases	
	Discussion of causality criteria	

Grid Type 5: Case Series

Domain	Criterion	Eliminatory
Study scheme	*Clear formulation of the objective*	✓
	Description of the disease studied	✓
	Representative sample	
Procedure of selection	*Inclusion criteria described and adequate*	✓
	Exclusion criteria described and adequate	
	Description of the type of recruitment	
	Indication of the number of refusals before the study	
	Identical procedures in all groups	
	Ethical rules respected	✓
Conduct of the study	*Standardization efforts*	✓
	Withdrawals indicated, explained and reasonable	
	Balanced withdrawals between groups	
	Results analysis	✓
	Detailed clinical description	
	Statistical tests only if representative	
	Verifiable raw data results	
Other elements	Discussion of selection and measurement biases	
	Absence of causality because not comparative	
	Absence of extrapolation if not representative	

Grid Type 6: Literature Review / Systematic Review

Domain	Criterion	Eliminatory
Study scheme	*Clear formulation of the objective*	✓
	A priori assumption	
	Clear definition and objective of the scope	✓
	Critical, qualitative or quantitative review	
Procedure of selection	*Inclusion criteria described and adequate*	✓
	Exclusion criteria described and adequate	
	Description and relevance of bibliographic databases	
	Relevance of study selection criteria	
	Taking into account unpublished literature	
	Indication of the number of exclusions before the study	
	Ethical rules respected	✓
Conduct of the study	Acceptable critical reading methods	
	Hidden Authors and Journal of Critical Readers	
	Double evaluation of the quality of studies	
	Explicit methods of information synthesis	
Results analysis	Description of selection results	
	Description of selected items	
	Taking into account the quality of studies	
	Analysis of the robustness of the conclusions	
	Publication bias analysis	

6A.5 Reading Grid Data Model

Reading Grid Assessment Record:

Field	Description
id	UUID
article_id	Parent article
soa_analysis_id	Parent SOA analysis
grid_type	rct | non_randomized_minors | cohort | case_control | case_series | literature_review
header_data	JSON — all common header fields (pre-populated + manual)
criteria_scores	JSON array — each criterion with: key, score (Y/I/N/NA/DNK or 0/1/2 for MINORS), comment
has_eliminatory_failure	boolean — computed: true if any eliminatory criterion scored N
overall_quality	select: very_good | pretty_good | weak_acceptable | unacceptable | not_qualified
overall_quality_auto	computed suggestion based on scores (user can override)
assessor	User who completed the assessment
assessed_at	Timestamp
status	pending | ai_drafted | reviewed | final

Auto-suggestion logic for overall quality:

- If `has_eliminatory_failure` = true → suggest `unacceptable`
- If MINORS: score 20-24 → `very_good`, 15-19 → `pretty_good`, 10-14 → `weak_acceptable`, <10 → `unacceptable`
- For Y/I/N grids: % of Y answers among applicable criteria → mapped to quality levels (thresholds configurable)
- User always has the final say — auto-suggestion is a starting point, not a constraint

6A.6 AI-Assisted Quality Assessment

For both Level 1 (QUADAS-2) and Level 2 (Reading Grids):

- The LLM can pre-fill all assessments based on the article's full text and abstract
- Each answer includes the AI's reasoning and source quote
- The user reviews, confirms, or overrides each answer
- Confidence indicators highlight where the AI is uncertain (e.g., "Unclear — the article does not explicitly state whether readers were blinded")

- Async batch processing for all articles, cancellable

For the common header: fields are pre-populated from the extraction grid where a mapping exists (see 6A.4 table). The AI fills remaining fields from the full text.

For reading grid criteria: the AI scores each criterion and provides a justification comment. Eliminatory criteria are flagged with a warning when the AI assigns N.

6A.7 Quality Score Integration

The quality assessment feeds into:

- **Benchmark weighting:** higher-quality studies can receive more weight in pooled calculations (configurable — user chooses equal weighting vs. quality-adjusted weighting)
- **Sensitivity analysis:** "What would the benchmark be if only high-quality studies were included?"
- **CER narrative:** the quality assessment summary is a required section in the CER's clinical evaluation methodology chapter
- **SOA section drafting:** the AI references quality scores when drafting section narratives ("Five high-quality studies (QUADAS-2: low risk of bias) consistently report sensitivity above 85%...")
- **Data contribution classification:** quality assessment informs the contribution level (see 6B)
- **Eliminatory filtering:** articles rated unacceptable via reading grid eliminatory criteria are flagged — user must decide whether to exclude from benchmarks or justify retention

---

6B. Data Contribution Classification (MEDDEV §9 Compliance)

6B.1 Purpose

MEDDEV 2.7/1 Rev 4 §9 requires assessing how each piece of clinical data contributes to the overall evidence base. Not all articles carry equal weight — some are pivotal evidence, others are supporting context.

6B.2 Classification

Each included article receives a data contribution level:

Level	Definition	Implications
pivotal	Directly applicable, high-quality evidence that the clinical evaluation relies upon. Performance data from comparable devices, populations, and endpoints.	Full QUADAS-2 + Reading Grid required. Referenced in claims. Must be discussed in CER.
supportive	Relevant evidence that supports the overall conclusions but is not directly relied upon. Different population, different endpoint, indirect comparison.	QUADAS-2 + Reading Grid recommended. May be referenced in background sections.
background	Contextual information (epidemiology, clinical pathway, guidelines). Not used for performance claims or benchmarks.	Quality assessment optional. Referenced in clinical SOA context sections.

6B.3 Assignment

- AI suggests the contribution level based on: SOA type, article's extraction grid data (especially `pertinence_type`, `relevance_to_intended_purpose`, `pivotal_aspects_covered`), quality assessment, how closely the article matches the device's intended use
- User confirms or overrides — with documented justification if upgrading or downgrading
- Validation rule: all articles contributing to benchmarks must be classified as `pivotal` and must have a completed quality assessment (both levels)

6B.4 Reporting

The CER must include a summary:

- Total included articles: N
- Pivotal: N (with full quality assessment summary)
- Supportive: N
- Background: N
- Distribution of quality ratings across pivotal studies
- Distribution of reading grid overall quality conclusions

---

7. Similar Device Registry (Clinical SOA — Section 6)

7.1 Purpose

During the Clinical SOA, articles frequently mention existing devices or software. The Similar Device Registry is a structured list built progressively during extraction and review, capturing all devices encountered in the literature.

7.2 Device Record

Field	Description
device_name	Product name
manufacturer	Company
type	Software, Hardware, Combined
regulatory_status	CE marked, FDA cleared, both, none, unknown
regulatory_reference	K-number, CE certificate ref
technology	DL, ML, rule-based, etc.
indications	Cleared/intended indications
source_article_ids	Which SLS articles mention this device
notes	Free-form

7.3 Auto-Population

When the grid column `devices_mentioned` is filled during extraction, the system auto-suggests adding new devices to the registry if they're not already present. The user confirms or dismisses.

7.4 Selection for Device SOA

From this registry, the user selects 1-2 devices as the "most similar devices" for equivalence analysis. This selection:

- Requires a justification (free text: why these devices were chosen as equivalent/similar)
- Defines the scope of the Device SOA
- Is referenced in the CER's equivalence argumentation
- Is logged in the audit trail
- Can be updated (with documented reason) if new information emerges

---

8. Benchmarks (Device SOA)

8.1 Purpose

Benchmarks are pooled statistical calculations from the device SOA extraction grids. They establish the State of the Art performance level that the device under evaluation must meet or exceed.

8.2 Calculation

For each poolable column in the device SOA grid:

1. Collect values from all articles where the value is reported (not null, not "not reported")
2. Weighted average (weighted by `dataset_size`):

```
Benchmark = Σ(value_i × n_i) / Σ(n_i)
```

3. Confidence interval (95% CI) calculated from the distribution
4. Statistics: number of studies, total subjects, min, max, median, standard deviation, I^2 heterogeneity index
5. Result stored as a benchmark record with full traceability to source articles

8.3 Selective Inclusion/Exclusion

The user can exclude specific articles from a benchmark calculation without excluding them from the SOA analysis itself.

When excluding an article from a benchmark:

- A reason is required (free text)
- The exclusion is logged in the audit trail
- The benchmark is recalculated automatically
- The benchmark record shows: "Calculated from N articles. M articles excluded: [list with reasons]"
- A sensitivity analysis is displayed: "Benchmark with all articles included: X.X% vs. current: Y.Y%"

8.4 Benchmark Record

Field	Description
metric_name	e.g., "Sensitivity"
pooled_value	Weighted average
ci_lower, ci_upper	95% CI bounds
n_studies	Number of studies in calculation
n_excluded	Number excluded (with reasons)
total_subjects	Sum of dataset_size across included studies
min, max, median, std_dev	Distribution stats
i_squared	Heterogeneity index (0-100%)

Field	Description
sensitivity_analysis_value	Pooled value if all articles included
quality_adjusted_value	Pooled value using quality-weighted calculation (optional)
source_grid_ids	Contributing extraction grid rows
excluded_grid_ids	Excluded rows with reasons
claim_text	Auto-generated claim statement

8.5 Auto-Generated Claims

For each benchmark, the system generates a claim statement:

> "The State of the Art sensitivity for AI-based CT perfusion analysis in acute ischemic stroke detection is 87.3% (95% CI: 82.1–92.5%), based on 12 studies encompassing 4,521 subjects. Heterogeneity was moderate (I^2 = 42%)."

Claims are auto-generated but editable. Changes tracked. Each claim is linked to its parent section in the Device SOA (`performance.claims`).

---

9. Comparison Tables (Device SOA)

9.1 Purpose

A side-by-side comparison of the device under evaluation against the similar devices identified in the Clinical SOA. This is a key CER deliverable for MDR equivalence argumentation.

9.2 Structure

Criterion	Your Device	Similar Device A	Similar Device B	SOA Benchmark
Intended Use	...	...	...	—
Technology	...	...	...	—
Regulatory Status	...	...	...	—
Dataset Size	...	...	...	Median: N
Sensitivity	(from Validation)	88.1%	...	87.3% [82.1–92.5]
Specificity	(from Validation)	92.3%	...	91.2% [88.0–94.4]
AUC	(from Validation)	0.95	...	0.94 [0.91–0.97]

Criterion	Your Device	Similar Device A	Similar Device B	SOA Benchmark
Clinical Benefit	(from Validation)	...	...	—
Known Risks	...	...	...	—

9.3 Features

- Auto-populated from extraction grids for similar devices
- Your device column filled from validation results (Module C) when available — empty initially
- SOA Benchmark column pulled from benchmark calculations
- Color coding: green if meets/exceeds benchmark, red if below, amber if within CI overlap
- Editable: user can add rows, edit values, add footnotes
- Exportable as DOCX table or XLSX

---

10. Digests (All SOA Types)

10.1 Purpose

AI-generated structured summaries for every included article. These support the narrative content of the thematic sections — each article's contribution to the evidence base is documented.

10.2 Generation

- Trigger: "Generate Digests" (batch for all, or single article)
- Input: article metadata + abstract + full-text PDF (if available) + extraction grid data
- Output: structured text paragraph (format TBD — initial structured template, will evolve with feedback)
- Async with progress for batch generation

10.3 Review

- Generated and editable — no formal approval workflow
- Rich text editor for editing
- Regenerate button available
- Edit history tracked (lightweight)

10.4 Status

Status	Meaning
not_generated	No digest yet

Status	Meaning
generated	AI-generated, not reviewed
edited	User has modified the generated text

---

11. SOA Analysis Workflow & Statuses

11.1 Lifecycle

draft → extraction → review → completed → locked

Status	What Happens	What's Editable
draft	Configuration: link SLS, choose/customize templates, configure similar devices	Everything
extraction	AI extraction running or in progress. Quality assessments can begin.	Grid data, template (add columns), reading grids
review	Human reviewing grids, generating benchmarks, drafting sections, writing claims, completing quality assessments	Grid data, benchmarks, sections, claims, digests, quality assessments
completed	All sections finalized, outputs ready for downstream modules	Minor edits only (logged)
locked	Frozen for a specific CER version. Read-only.	Nothing

11.2 Lock Behavior

- Locking is triggered when the SOA is included in a CER version
- A locked SOA can be duplicated to create a new version for updates
- Unlocking requires manager role and a documented justification

---

12. Exports & Outputs

Export	Format	Content
Extraction Grid	XLSX	Full spreadsheet: articles × columns, with metadata and appraisal fields

Export	Format	Content
Quality Assessment — Regulatory (QUADAS-2)	XLSX / DOCX	QUADAS-2 (or alternative) scores per article, domain-level detail
Quality Assessment — Internal (Reading Grids)	XLSX / DOCX	Per-article reading grid with all criteria scores, comments, overall quality, eliminatory flags
Quality Assessment — Combined Summary	XLSX / DOCX	Both levels side-by-side per article: QUADAS-2 overall + Reading Grid overall + data contribution level
Data Contribution Summary	XLSX	All articles with contribution level (pivotal/supportive/background) and quality ratings (both levels)
Benchmarks Summary	XLSX	All benchmarks with stats, sources, exclusions, quality-adjusted variants
Comparison Table	DOCX / XLSX	Side-by-side device comparison
Article Digests	DOCX	All digests formatted as narrative
Claims List	XLSX / DOCX	All claims with type, text, article references
SOA Sections	DOCX	All thematic sections assembled as a document
Full SOA Report	DOCX	Complete report: scope + methodology + quality assessment summary + sections + grids + benchmarks + claims + comparison table

These exports feed into the CER assembly (Module E) but can also be used standalone.

---

13. Async Operations

Operation	Estimated Duration	Async?
AI extraction (batch)	5-30s per article × N	Yes
AI extraction (single)	5-30s	No (inline with spinner)
Column fill	3-10s per article × N	Yes
Quality assessment — QUADAS-2 (batch)	5-15s per article × N	Yes
Quality assessment — Reading Grid (batch)	10-30s per article × N	Yes
Quality assessment (single, either level)	5-30s	No (inline with spinner)

Operation	Estimated Duration	Async?
Benchmark calculation	< 2s	No (inline)
Digest generation (batch)	3-10s per article × N	Yes
Section drafting (single)	10-30s	No (inline with spinner)
Section drafting (all)	10-30s × N sections	Yes
Report export (DOCX)	5-15s	No (inline with spinner)

All async operations: progress bar, cancellable, items preserved on cancel.

---

14. Permissions

Role	Can Do
clinical / regulatory	Full access: create analyses, extract, review, generate benchmarks, draft sections, complete quality assessments
manager	Full access + lock/unlock with justification
collaborateur	Read-only access to completed/locked analyses and exports
admin	Everything + manage default grid/section/reading grid templates at system level

---

15. Open Items

1. **Digest format:** TBD — initial structured template, will evolve during implementation.
2. **Meta-analysis rigor:** Current spec uses weighted averages + I^2. If full random-effects meta-analysis is needed later, the benchmark calculation is modular and upgradeable.
3. **Product-specific grid columns:** For specific domains (CT perfusion: CBF, CBV, MTT, Tmax), added as custom columns. A library of domain-specific template presets can be built over time.
4. **MAUDE / EUDAMED integration:** Section 4.4 of the Device SOA references external safety databases. For v1 this is manual import. API integration can be added later.
5. **Section-to-grid mapping:** The `section_mapping` on grid columns creates a loose coupling between extraction data and narrative sections. This helps AI drafting but is not a hard dependency.
6. **QUADAS-2 customization:** The adapted QUADAS-2 form covers standard diagnostic accuracy studies. The system auto-suggests the tool based on `study_design`, but the user can override.

7. **Quality-adjusted benchmarking:** The spec supports optional quality-adjusted weighting in benchmark calculations. The weighting formula (e.g., quality score × sample size) needs to be validated with clinical affairs to ensure it's methodologically sound and defensible to a Notified Body.
8. **Equivalence contractual access:** For most Avicenna.AI products, equivalence is based on public data only (no manufacturer contracts). The spec includes a documentation template for this.
9. **Reading grid template versioning:** If the internal reading grid criteria evolve (e.g., new HAS guidelines), existing completed assessments retain their original grid version. New assessments use the updated template. Both versions are archived.
10. **Cross-referencing between Level 1 and Level 2 quality assessments:** When QUADAS-2 and Reading Grid assessments disagree (e.g., QUADAS-2 says "low risk" but Reading Grid says "unacceptable" due to an eliminatory criterion), the system flags the discrepancy for manual resolution. The user documents which assessment takes precedence and why.
11. **Reading grid AI accuracy:** The reading grids have ~25 criteria each with nuanced judgment calls. AI accuracy is expected to be lower than for QUADAS-2 (which has fewer, more standardized questions). Initial implementation should default to DNK when the AI confidence is below a threshold, rather than risking incorrect scores. The threshold is configurable.